

BASIC ELECTRICAL AND ELECTRONIC ENGINEERING

Course Code: 2501EE01

Comprehensive Question & Answer Guide

Units 1, 2 & 3 — Detailed Answers with Diagrams

Unit 1	Basic Electrical Circuits & AC Fundamentals
Unit 2	Electrical Machines & Measuring Instruments
Unit 3	Power Systems, Safety & Energy Management

Department of Freshman Engineering

UNIT 1: BASIC ELECTRICAL CIRCUITS & AC FUNDAMENTALS

Question 1

Q1a. Find the Equivalent Resistance & Total Current in a Circuit

Series Circuit

When resistors are connected end-to-end, the same current flows through all of them. The equivalent resistance is simply the sum of all individual resistances.

$$R_{eq} \text{ (series)} = R_1 + R_2 + R_3 + \dots + R_n$$

Example: $R_1 = 4\Omega$, $R_2 = 6\Omega$, $V = 20V \rightarrow R_{eq} = 4+6 = 10\Omega \rightarrow I = V/R_{eq} = 20/10 = 2 \text{ A}$

Parallel Circuit

When resistors share the same two nodes, the same voltage appears across each. The reciprocal of the equivalent resistance equals the sum of reciprocals.

$$1/R_{eq} \text{ (parallel)} = 1/R_1 + 1/R_2 + 1/R_3$$

Example: $R_1 = 6\Omega$, $R_2 = 3\Omega$, $V = 12V \rightarrow 1/R_{eq} = 1/6+1/3 = 3/6 \rightarrow R_{eq} = 2\Omega \rightarrow I_{total} = 6 \text{ A}$

Equivalent Resistance — Series & Parallel

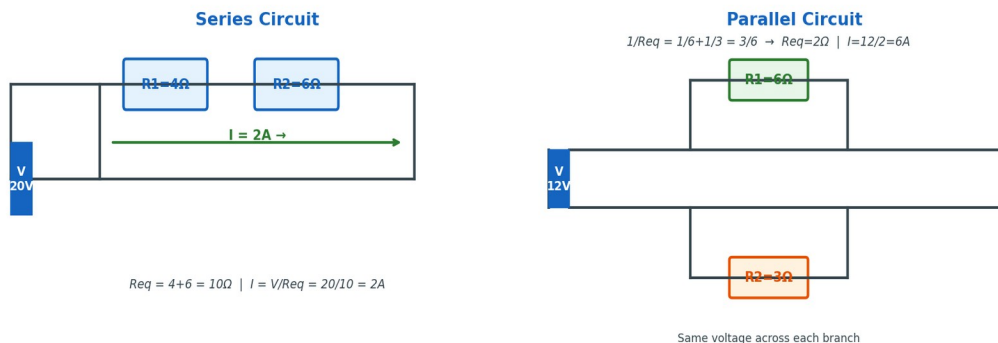


Fig 1.1 — Series & Parallel Circuit with Calculations

NOTE: In a series circuit, current is constant; in a parallel circuit, voltage is constant.

Q1b. What is Power Factor? What is the Significance?

Definition

Power Factor (PF) is the cosine of the phase angle (ϕ) between the voltage and current waveforms in an AC circuit. It indicates how efficiently the electrical power is being converted to useful work.

$$\text{Power Factor} = \cos \phi = \text{Active Power (P)} / \text{Apparent Power (S)} = P / (V \times I)$$

Types of Power Factor

- **Unity PF ($\cos \phi = 1$):** Pure resistive load — voltage and current are in phase. Maximum efficiency.
- **Lagging PF ($\cos \phi < 1$, lagging):** Inductive load (motors, coils) — current lags voltage. Most common in industry.
- **Leading PF ($\cos \phi < 1$, leading):** Capacitive load (capacitors) — current leads voltage.

Power Triangle

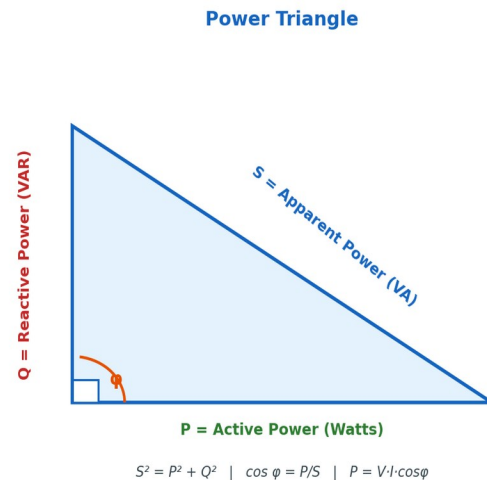


Fig 1.2 — Power Triangle

- **Active Power (P):** Measured in Watts (W). Does useful work — heat, light, mechanical motion.
- **Reactive Power (Q):** Measured in VAR. Stored in inductors/capacitors & returned to source each cycle.
- **Apparent Power (S):** Measured in VA. Total power supplied: $S = V \times I$.

$$S^2 = P^2 + Q^2 \quad | \quad P = S \times \cos \phi \quad | \quad Q = S \times \sin \phi \quad | \quad \cos \phi = P/S$$

Significance of Power Factor

- Low PF means higher current is needed for the same useful power — increases conductor size & cost.
- Utility companies charge a penalty for PF below 0.85–0.90.
- Low PF causes more I^2R losses in transmission lines, increasing energy cost.
- PF correction is achieved using capacitor banks in parallel with inductive loads.

 **NOTE:** Unity power factor ($\cos \phi = 1$) is ideal. Most industrial loads have a lagging PF due to induction motors.

Question 2

Q2a. Verify Kirchhoff's Current Law (KCL) for a Given Circuit

Statement of KCL

"The algebraic sum of all currents meeting at a node (junction) is equal to zero." KCL is based on the conservation of electric charge — charge can neither be created nor destroyed at a node.

$$\Sigma I_{in} = \Sigma I_{out} \quad \text{OR} \quad \Sigma I = 0 \text{ at any node}$$

Sign Convention

- Currents entering a node → considered positive (+)
- Currents leaving a node → considered negative (-)

Numerical Verification

At node A: $I_1 = 5\text{A}$ (entering), $I_2 = 2\text{A}$ (leaving), $I_3 = 3\text{A}$ (leaving).

$$\text{Sum} = I_1 - I_2 - I_3 = 5 - 2 - 3 = 0 \quad \checkmark \quad \text{KCL Verified}$$

Kirchhoff's Current Law (KCL)

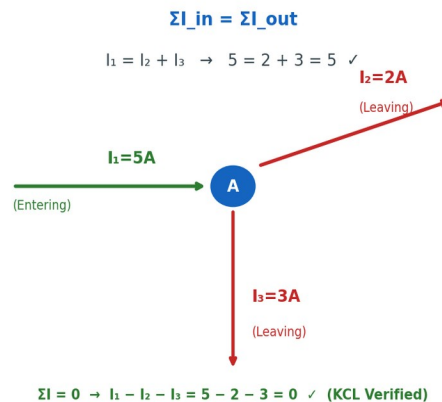


Fig 2.1 — KCL Verification at Node A

NOTE: KCL applies to any node — a point where two or more circuit elements are connected together.

Q2b. Define KCL & KVL and Explain with Numerical Examples

Kirchhoff's Current Law (KCL)

At any node in a circuit, the total current entering equals the total current leaving.

$$I_1 + I_2 = I_3 + I_4 \quad (\text{at a node})$$

Kirchhoff's Voltage Law (KVL)

In any closed loop in a circuit, the algebraic sum of all voltages (EMFs and drops) is zero.

$$\Sigma V = 0 \quad \text{in a closed loop}$$

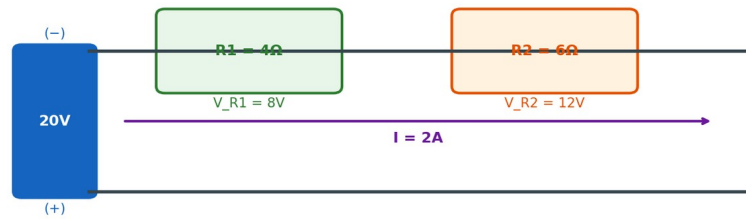
KVL Numerical Example

Series circuit: $V = 20\text{V}$, $R_1 = 4\Omega$, $R_2 = 6\Omega$.

- $I = V / (R_1 + R_2) = 20 / 10 = 2\text{ A}$
- $V_{R1} = I \times R_1 = 2 \times 4 = 8\text{ V}$
- $V_{R2} = I \times R_2 = 2 \times 6 = 12\text{ V}$

KVL check: $20 - 8 - 12 = 0$ ✓ KVL Verified

Kirchhoff's Voltage Law (KVL) — Series Circuit



$$\text{KVL: } V - V_{R1} - V_{R2} = 0 \rightarrow 20 - 8 - 12 = 0 \quad \checkmark$$

$$I = V/(R1 + R2) = 20/10 = 2A$$

Fig 2.2 — KVL Series Circuit Verification

Question 3

Q3a. Define Ohm's Law and What Are the Limitations?

Definition

Ohm's Law states that the current flowing through a conductor is directly proportional to the voltage applied across it, provided the temperature and other physical conditions remain constant.

$$V = I \times R \quad (\text{Volts} = \text{Amperes} \times \text{Ohms})$$

Where: V = Voltage (Volts), I = Current (Amperes), R = Resistance (Ohms, Ω)

Limitations of Ohm's Law

- **Non-linear devices:** Diodes, transistors, and thyristors do NOT obey Ohm's law — their V-I relationship is non-linear.
- **Temperature dependence:** Resistance changes with temperature. Ohm's law holds only at constant temperature.
- **Electrolytes:** Solutions like acid/salt water have complex V-I relationships.
- **Gaseous conductors:** In gas discharge tubes, the V-I relationship is non-linear.
- **Semiconductors:** At high voltages, current saturates or breaks down — not linear.
- **Unilateral elements:** Diodes allow current only in one direction and do not follow Ohm's law symmetrically.

NOTE: Ohm's Law applies only to ohmic (linear) conductors under constant physical conditions.

Q3b. Compare AC & DC Supply

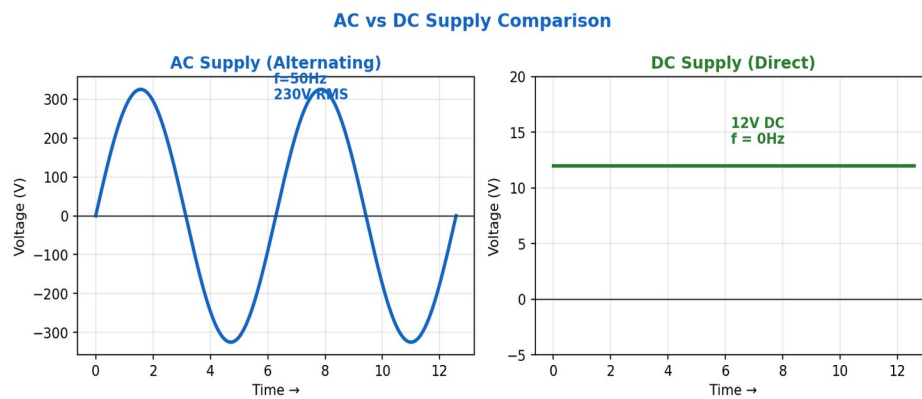


Fig 3.1 — AC vs DC Waveforms

Parameter	AC Supply	DC Supply
Full Form	Alternating Current	Direct Current
Direction	Reverses periodically	Flows in one direction only
Frequency	50 Hz (India) / 60 Hz (USA)	0 Hz (no oscillation)
Generation	Alternators	Batteries, rectifiers, solar cells
Transformer Use	Can step up/down easily	Transformer cannot be used
Transmission	Efficient over long distances	Difficult over long distances
Voltage	230V single-phase, 415V three-phase	Fixed (5V, 12V, 24V, 48V)
Applications	Household, industrial supply	Electronics, charging, traction

Question 4

Q4a. Find I_x in a Circuit (Mesh/Node Analysis using KVL)

Method: Mesh Analysis (KVL)

To find an unknown current I_x using KVL in a circuit with voltage sources and resistors:

- Step 1: Assign mesh currents I_1, I_2, I_3 in each loop (clockwise preferred).
- Step 2: Write KVL equation for each loop — sum of voltage drops = EMF.
- Step 3: Solve the simultaneous equations for each mesh current.
- Step 4: I_x = difference of mesh currents through the shared branch.

$$\text{Loop 1: } V - I_1 \cdot R_1 - (I_1 - I_2) \cdot R_2 = 0$$

$$\text{Loop 2: } (I_2 - I_1) \cdot R_2 - I_2 \cdot R_3 = 0$$

Solving these simultaneously gives I_1 and I_2 . Then $I_x = I_1 - I_2$ (current through shared branch).

NOTE: Always assign mesh current direction (clockwise preferred) consistently before writing KVL equations.

Q4b. Define: Time Period, Frequency, Amplitude, Peak-to-Peak Value

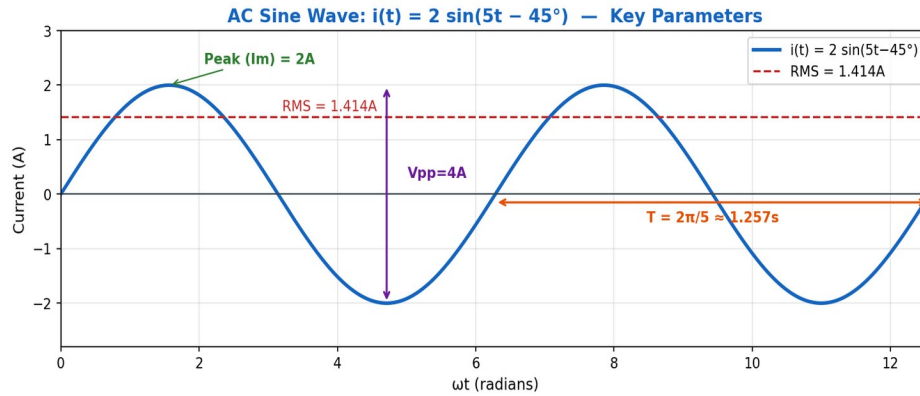


Fig 4.1 — AC Sine Wave — Key Parameters

(i) **Time Period (T):** Time taken to complete one full cycle of an AC waveform. Unit: seconds (s).

$$T = 1/f$$

(ii) **Frequency (f):** Number of complete cycles per second. Unit: Hertz (Hz). In India, $f = 50$ Hz.

$$f = 1/T$$

(iii) **Amplitude (Peak Value, V_m or I_m):** Maximum value of the AC waveform from the zero (mean) line — the highest point of the sine wave.

$$v(t) = V_m \sin(\omega t) \rightarrow V_m \text{ is the amplitude}$$

(iv) **Peak-to-Peak Value (V_{pp}):** Total vertical distance from negative peak to positive peak.

$$V_{pp} = 2 \times V_m \quad (\text{twice the amplitude})$$

Question 5

Q5a. $i(t) = 2 \sin(5t - 45^\circ)$ — Find All Parameters

Given: $i(t) = 2 \sin(5t - 45^\circ) \rightarrow$ Standard form: $i(t) = I_m \sin(\omega t + \phi)$

Parameter	Formula	Calculation	Result
Peak Value (I_m)	$I_m = \text{coefficient}$	$I_m = 2$	2 A
Angular Freq. (ω)	$\omega = \text{coefficient of } t$	$\omega = 5$	5 rad/s
Frequency (f)	$f = \omega / 2\pi$	$5 / (2\pi)$	≈ 0.796 Hz
Time Period (T)	$T = 2\pi/\omega = 1/f$	$2\pi/5$	≈ 1.257 s
RMS Value	$I_{rms} = I_m / \sqrt{2}$	$2 / \sqrt{2}$	≈ 1.414 A
Average Value	$I_{avg} = (2/\pi) \times I_m$	$(2/\pi) \times 2$	≈ 1.273 A
Phase Angle (ϕ)	$\phi = \text{angle in expression}$	-45°	-45° (lags reference)

NOTE: RMS value is what AC meters display. It is the DC equivalent heating value of AC.

Q5b. Explain Power Triangle and Define All Terms

The Power Triangle

When AC power is drawn by a load, it can be represented as a right-angled triangle with three sides representing three types of power:

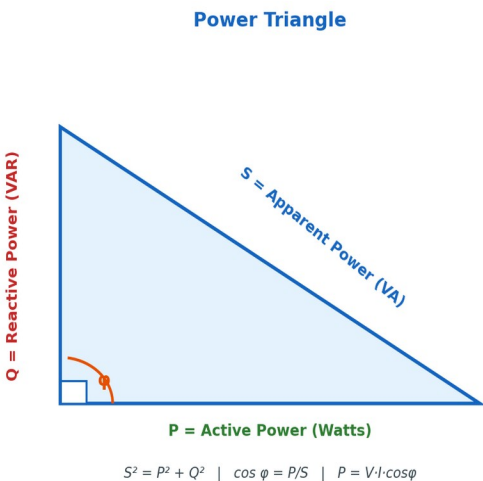


Fig 5.1 — Power Triangle

- **Active Power (P):** Real power consumed and converted to useful work (heat, light, mechanical). $P = V \cdot I \cdot \cos \phi$ [Watts]
- **Reactive Power (Q):** Power stored in inductors/capacitors and returned to source each cycle. Does no useful work. $Q = V \cdot I \cdot \sin \phi$ [VAR]
- **Apparent Power (S):** Total power supplied by source $= V \times I$. $S = \sqrt{(P^2 + Q^2)}$ [VA]
- **Power Factor (cosφ):** Ratio of active power to apparent power. $\cos \phi = P/S$. Indicates efficiency of power use.

$S^2 = P^2 + Q^2 \quad | \quad \cos \phi = P/S \quad | \quad \sin \phi = Q/S \quad | \quad \tan \phi = Q/P$

Question 6

Q6a. $v(t) = 10 \cos(4t + 45^\circ)$ — Find All Parameters

Conversion: cosine to sine form

$\cos(\theta) = \sin(\theta + 90^\circ) \rightarrow v(t) = 10 \sin(4t + 45^\circ + 90^\circ) = 10 \sin(4t + 135^\circ)$

Parameter	Formula	Calculation	Result
Peak Value (V_m)	$V_m = \text{coefficient}$	$V_m = 10$	10 V
Angular Freq. (ω)	$\omega = \text{coefficient of } t$	$\omega = 4$	4 rad/s
Frequency (f)	$f = \omega / 2\pi$	$4 / (2\pi)$	≈ 0.637 Hz
Time Period (T)	$T = 2\pi/\omega$	$2\pi/4 = \pi/2$	≈ 1.571 s

Parameter	Formula	Calculation	Result
RMS Value	$V_{rms} = V_m / \sqrt{2}$	$10 / \sqrt{2}$	$\approx 7.07 \text{ V}$
Average Value	$V_{avg} = (2/\pi) \times V_m$	$(2/\pi) \times 10$	$\approx 6.37 \text{ V}$
Phase Angle (ϕ)	$\phi = \text{angle in sine form}$	$+135^\circ$ (or $+45^\circ$ in cos)	Leads reference by 45°

Q6b. Define & Derive RMS Value and Average Value of AC

RMS (Root Mean Square) Value

The RMS value of AC is the DC equivalent value that produces the same heating effect in the same resistance over the same time period.

Derivation of RMS Value

For $v(t) = V_m \sin(\omega t)$, over one full cycle (0 to 2π):

$$\begin{aligned}
 V_{rms} &= \sqrt{\left[\frac{1}{2\pi} \times \int_0^{2\pi} V_m^2 \sin^2(\omega t) d(\omega t) \right]} \\
 &= \sqrt{\left[\frac{V_m^2}{2\pi} \times \int_0^{2\pi} \frac{(1 - \cos 2\omega t)}{2} d(\omega t) \right]} \\
 &= \sqrt{\left[\frac{V_m^2}{2} \right]} = \frac{V_m}{\sqrt{2}} \approx 0.707 \times V_m
 \end{aligned}$$

Average Value of AC

The average value of AC is the DC equivalent equal to the mean of all instantaneous values over a half-cycle. (Full cycle average = 0 for pure sine wave, so half-cycle is used.)

$$\begin{aligned}
 V_{avg} &= \frac{1}{\pi} \times \int_0^{\pi} V_m \sin(\omega t) d(\omega t) \\
 &= \frac{2}{\pi} \times V_m \approx 0.637 \times V_m
 \end{aligned}$$

 **NOTE:** For a pure sine wave: $V_{rms} = 0.707 V_m$ | $V_{avg} = 0.637 V_m$ | Form Factor = $V_{rms}/V_{avg} = 1.11$

UNIT 2: ELECTRICAL MACHINES & MEASURING INSTRUMENTS

Question 1

Q1a. What is an Electrical Machine? Explain DC Generator Operation

Definition of Electrical Machine

An electrical machine is a device that converts electrical energy to mechanical energy or vice versa. Examples: DC motor, DC generator, induction motor, transformer (static machine).

Operating Principle of DC Generator

A DC generator converts mechanical energy into electrical energy based on Faraday's Law of Electromagnetic Induction: when a conductor moves through a magnetic field, an EMF is induced in it.

$$\text{Induced EMF: } e = B \times L \times v \times \sin\theta$$

Main Parts of a DC Generator

- **Stator (Field System):** Produces the magnetic field via field windings or permanent magnets.
- **Rotor (Armature):** Rotating part that carries armature winding where EMF is induced.
- **Commutator:** Converts alternating EMF induced in armature into DC output — acts as mechanical rectifier.
- **Brushes:** Stationary carbon contacts that collect current from the commutator.
- **Yoke:** Outer iron frame providing mechanical support and completing the magnetic circuit.

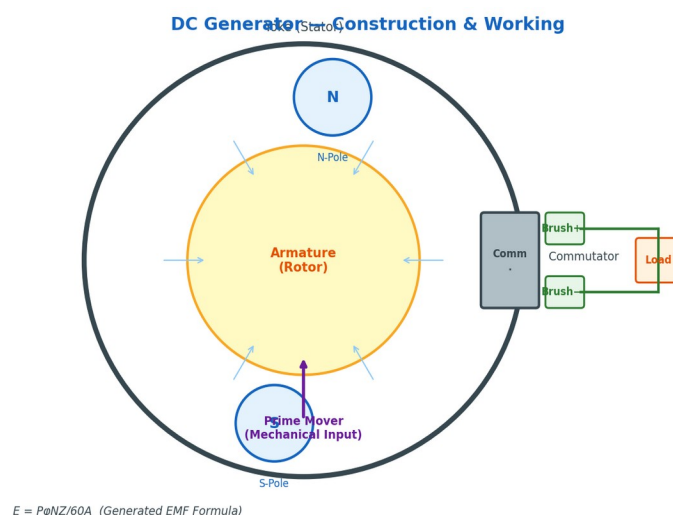


Fig 1.1 — DC Generator Construction & Working

Generated EMF Formula

$$E = \frac{(P \times \phi \times N \times Z)}{(60 \times A)} \text{ Volts}$$

Where: P=poles, ϕ =flux/pole (Wb), N=speed (rpm), Z=total conductors, A=parallel paths

Q1b. Explain the Operation of Megger with Neat Diagram

Purpose of Megger

A Megger is an instrument used to measure very high resistances — particularly the insulation resistance of cables, motors, and transformers. It checks whether insulation is healthy (no leakage current).

Working Principle

A Megger uses a high DC voltage (typically 500V, 1000V, or 2500V) to test insulation. It is based on the ratio meter principle:

- Two coils (deflecting coil and control coil) are mounted at right angles on the same spindle inside the magnetic field.
- A hand-cranked or electronic generator produces the high test voltage.
- The deflecting coil carries current proportional to the test voltage.
- The control coil carries current proportional to the insulation resistance.
- The pointer deflects to indicate the resistance on the Mega-ohm (MΩ) scale.

Megger — Insulation Resistance Tester

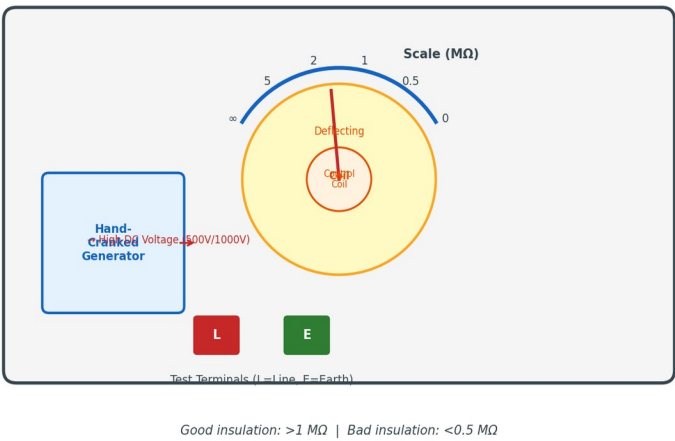


Fig 1.2 — Megger Construction & Working

Reading Interpretation

Reading	Insulation Condition	Action
> 100 MΩ	Excellent insulation	Safe to use
1 – 100 MΩ	Good insulation	Acceptable
0.5 – 1 MΩ	Doubtful insulation	Monitor closely
< 0.5 MΩ	Bad insulation	Replace immediately

 **NOTE:** Megger works at high voltage, so equipment being tested must be de-energized and disconnected from load.

Question 2

Q2a. Classify Measuring Instruments for AC & DC Quantities with Basic Principle

Classification by Working Principle

Instrument Type	Works On	Principle	Used For
PMMC	DC only	Interaction of permanent magnet & current-carrying coil	DC voltmeters, ammeters
Moving Iron (MI)	AC & DC	Magnetic attraction or repulsion of soft iron	AC voltmeters, ammeters
Electrodynamometer	AC & DC	Interaction of two magnetic fields	Wattmeters, PF meters
Rectifier Type	AC	AC → DC via rectifier, measured by PMMC	AC voltmeters, ammeters
Thermocouple	AC & DC	Heating effect of current (thermoelectric)	High-frequency measurements
Induction Type	AC only	Eddy current & flux interaction	Energy meters (kWh)

Classification by Quantity Measured

- **Ammeter:** Measures current. Connected in series. Very low internal resistance.
- **Voltmeter:** Measures voltage. Connected in parallel. Very high internal resistance.
- **Wattmeter:** Measures active power. Has current coil (series) and pressure coil (parallel).
- **Energy Meter:** Measures energy consumed over time (kWh). Induction disc type.
- **Multimeter:** Measures V, I, and R. Can be analog or digital.

Q2b. Classify Resistors & Explain High Resistance Measurement

Classification of Resistors

- **Fixed Resistors:** Carbon Composition, Carbon Film, Metal Film, Wire-Wound — fixed resistance value.
- **Variable Resistors:** Potentiometers (3-terminal) and Rheostats (2-terminal) — manually adjustable.
- **Special Resistors:** LDR (Light Dependent), NTC/PTC Thermistors (temperature sensitive), VDR (Voltage Dependent).

Measurement of High Resistance — Loss of Charge Method

Used when resistance is very high ($>100\text{ M}\Omega$) — e.g., insulation resistance of cables.

- Step 1: Charge a capacitor C to a known voltage V_0 .
- Step 2: Connect the unknown high resistance R in parallel with the charged capacitor.
- Step 3: The capacitor discharges through R. Measure voltage V after time t.

$$R = t / (C \times \ln(V_0/V))$$

Where: t = elapsed time, C = capacitance, V_0 = initial voltage, V = voltage after time t

 **NOTE:** Megger can also directly measure high resistance. The loss of charge method is accurate for very high ($\text{M}\Omega$ to $\text{G}\Omega$) resistances.

Question 3

Q3a. Difference between Back EMF & Generated EMF — Function of Each

Generated EMF (E_g) — in DC Generator

In a DC generator, the EMF generated in armature conductors due to rotation in a magnetic field.

$$E_g = (P \times \phi \times N \times Z) / (60 \times A) \quad \text{Volts}$$

Function: Source EMF that drives current through the external circuit (load).

Back EMF (E_b) — in DC Motor

When the armature of a DC motor rotates, it also acts as a generator and produces an EMF that opposes the applied voltage. This opposing EMF is called Back EMF or Counter EMF.

$$E_b = V - I_a \times R_a \quad (\text{in a DC motor})$$

Function: Regulates armature current and controls motor speed. Higher speed $\rightarrow$ Higher $E_b \rightarrow$ Less armature current.

Aspect	Back EMF (Motor)	Generated EMF (Generator)
Definition	EMF opposing applied voltage	EMF produced by rotating armature
Effect	Controls armature current	Drives current to external load
Variation	Increases with motor speed	Increases with armature speed
Formula	$E_b = V - I_a \times R_a$	$E_g = P\phi NZ/60A$
Purpose	Speed regulation	Electrical power generation

Q3b. Why Are Single-Phase Induction Motors Not Self-Starting? Explain Two Types

Why Not Self-Starting?

A single-phase induction motor is NOT self-starting because:

- A single-phase supply produces a pulsating (alternating) magnetic field, NOT a rotating magnetic field.
- A pulsating field resolves into two equal and opposite rotating fields that cancel each other.
- At standstill, equal opposing fields produce zero resultant torque — motor cannot start on its own.
- Once given an initial rotation, it continues in that direction.

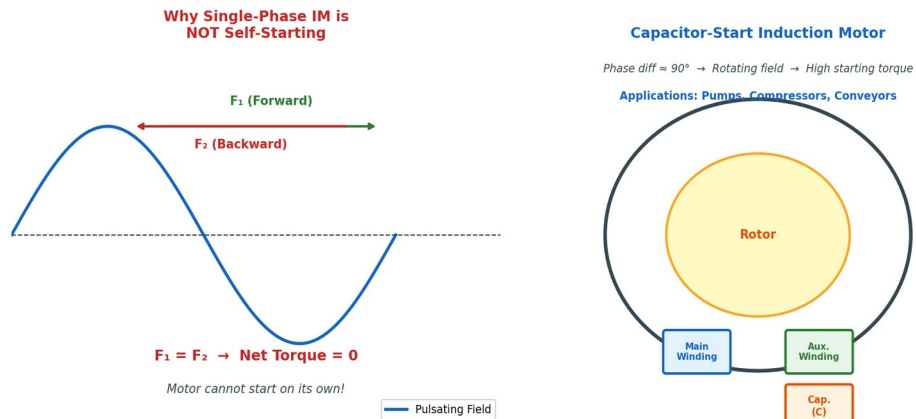


Fig 3.1 — Pulsating Field Analysis & Capacitor-Start Motor

Type 1: Capacitor-Start Induction Motor

- A capacitor is connected in series with the auxiliary winding to create a phase difference ($\sim 90^\circ$).
- Produces a strong starting torque (5–8 times full load torque).
- Capacitor is disconnected after starting using a centrifugal switch.
- Applications: Compressors, conveyors, pumps requiring high starting torque.

Type 2: Split-Phase Induction Motor

- Has two windings: main winding (high resistance, low inductance) and auxiliary winding (low resistance, high inductance).
- Auxiliary winding creates a phase displacement ($\sim 30^\circ$) to produce a rotating field.
- Applications: Fans, washing machines, refrigerators (small household appliances).

Type 3: Shaded Pole Motor

- Part of each stator pole is surrounded by a shading coil (copper ring).
- Creates a small phase shift ($\sim 30^\circ$) between shaded and unshaded portions.
- Very low starting torque — used for light loads: table fans, hairdryers.

Question 4

Q4a. Role of Transformer in Electrical Power System & Operation

Role of Transformer in Power System

- **Step-Up Transformer at generating station:** Steps up voltage (e.g., 11 kV → 220 kV) for long-distance transmission, reducing I^2R losses.
- **Step-Down Transformers at sub-stations:** 220 kV → 33 kV → 11 kV → 415V/230V for safe distribution to consumers.
- **Isolation:** Provides electrical isolation between primary and secondary circuits.
- **Impedance matching:** Used in electronics for maximum power transfer.

Operating Principle

A transformer works on the principle of Mutual Electromagnetic Induction. When alternating current flows through the primary winding, it creates an alternating magnetic flux in the core. This flux links with the secondary winding and induces an EMF in it.

Transformer — Construction & Working Principle

$$N_2/N_1 = V_2/V_1 = I_2/I_1 \quad | \quad \text{Works only on AC (Faraday's Law)}$$

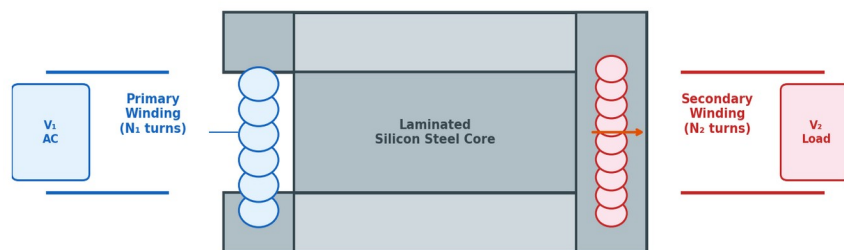


Fig 4.1 — Transformer Construction & Working

$$\text{Turns Ratio: } N_1/N_2 = V_1/V_2 = I_2/I_1$$

Main Parts

- **Primary Winding (N_1 turns):** Connected to input AC supply. Creates magnetic flux.
- **Secondary Winding (N_2 turns):** Delivers power to the load via mutual induction.
- **Core (Laminated Silicon Steel):** Provides low-reluctance magnetic path. Laminated to reduce eddy current losses.

$$\text{Step-up: } N_2 > N_1 \ (V_2 > V_1) \quad | \quad \text{Step-down: } N_2 < N_1 \ (V_2 < V_1)$$

🔧 **NOTE:** Transformer efficiency is very high (98–99%). It works ONLY on AC, not on DC.

Q4b. Find Direction of Mechanical & Electrical Output — Fleming's Rules

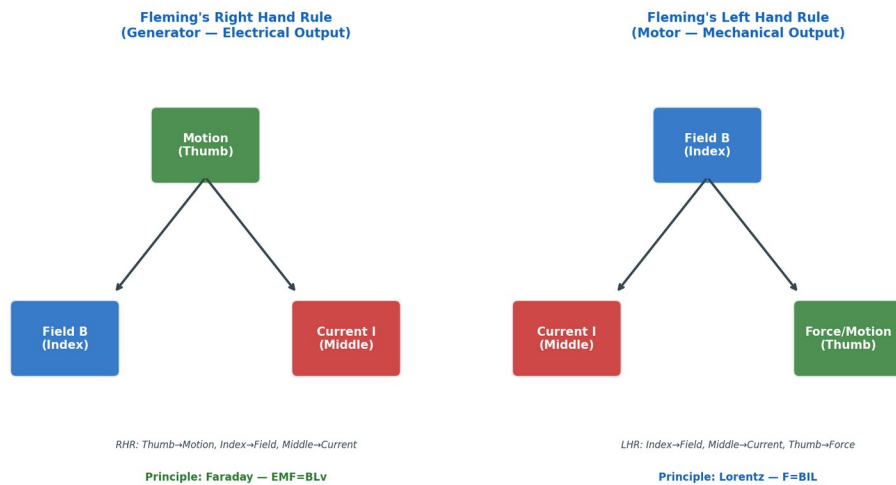


Fig 4.2 — Fleming's Right & Left Hand Rules

For Electrical Output (Generator) — Fleming's Right Hand Rule

- **Thumb:** → Direction of Motion of conductor (mechanical input)
- **Index Finger:** → Direction of Magnetic Field (B, North to South)
- **Middle Finger:** → Direction of Induced Current (electrical output)

For Mechanical Output (Motor) — Fleming's Left Hand Rule

- **Forefinger (Index):** → Direction of Magnetic Field (B)
- **Middle Finger:** → Direction of Current (I)
- **Thumb:** → Direction of Force/Motion (mechanical output)

Underlying Principle

- **Generator (Faraday's Law):** $EMF = B \times L \times v$ (conductor moves in magnetic field → EMF induced)
- **Motor (Lorentz Force):** $F = B \times I \times L$ (force on current-carrying conductor in magnetic field → rotation)

NOTE: Generator converts mechanical → electrical; Motor converts electrical → mechanical. Both use the same construction but different input/output.

Question 5

Q5a. Describe the Role of Commutator in DC Machines

What is a Commutator?

The commutator is a cylindrical assembly of copper segments separated by mica insulation, mounted on the armature shaft. Carbon brushes rest on it to collect current.

Role in DC Generator

- The armature coils generate alternating (AC) EMF as they rotate.

- The commutator acts as a mechanical rectifier — it reverses the connection of each coil to the external circuit just as the EMF in that coil reverses.
- Result: Current in the external circuit is always in the same direction → DC output.

Role in DC Motor

- The commutator reverses the current direction in each armature coil at the right instant.
- This ensures the torque on the armature is always in the same rotational direction.
- Without commutator, force would reverse every half turn — motor would oscillate rather than rotate.

🔧 **NOTE:** Commutator + Brushes together form the 'mechanical switch' that enables unidirectional current flow (generator) or continuous rotation (motor).

Q5b. Measurement of Medium Resistance — Wheatstone Bridge

Principle

The Wheatstone Bridge measures medium resistances (1Ω to $1M\Omega$) accurately using a null-deflection method — the bridge is said to be balanced when the galvanometer reads zero.

Wheatstone Bridge — Medium Resistance Measurement

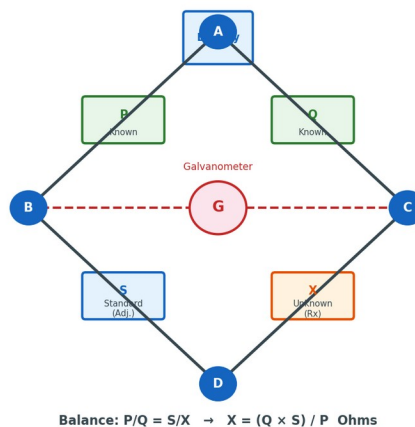


Fig 5.1 — Wheatstone Bridge

Circuit

The bridge consists of four resistances: P, Q (known ratio arms), S (standard adjustable resistance), and X (unknown resistance). A galvanometer G connects between junctions.

Derivation of Balance Condition

At balance (zero galvanometer deflection), applying KVL:

- Current through P and Q: I_1 | Current through S and X: I_2
- At balance: $I_1 \cdot P = I_2 \cdot S$ and $I_1 \cdot Q = I_2 \cdot X$

Dividing both equations:

$$P/Q = S/X \rightarrow X = (Q \times S) / P \text{ Ohms}$$

Procedure

- Step 1: Connect the unknown resistor X in one arm of the bridge.
- Step 2: Adjust the standard resistance S until galvanometer reads zero (balanced).
- Step 3: Apply formula $X = (Q/P) \times S$ to calculate unknown resistance.

NOTE: Wheatstone Bridge is not suitable for very low ($<1\Omega$) or very high ($>1M\Omega$) resistances due to galvanometer sensitivity limitations.

Question 6

Q6a. Draw and Explain the PMMC Instrument

PMMC — Permanent Magnet Moving Coil (D'Arsonval Movement)

Construction

- **Permanent Magnet:** Creates a strong, uniform magnetic field in the air gap. Made of Alnico or ferrite.
- **Moving Coil:** Rectangular coil of insulated copper wire wound on an aluminium former, pivoted in the magnetic field.
- **Former (Aluminium):** Provides eddy current damping — prevents oscillation.
- **Springs (Control Springs):** Two hairsprings provide restoring torque and carry current to the coil.
- **Pointer:** Attached to coil; deflects over calibrated scale.
- **Iron Core (Cylindrical):** Placed inside the coil to make the magnetic field radial and uniform.

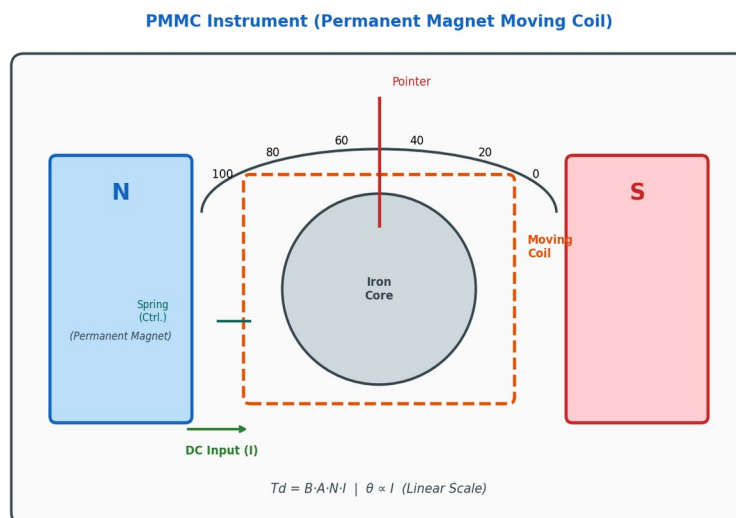


Fig 6.1 — PMMC Instrument Construction & Working

Working Principle

When DC passes through the moving coil, it experiences a force due to Lorentz interaction with the permanent magnet's field ($F = BIL$). The coil deflects until deflecting torque = controlling torque of the spring.

$$\text{Deflecting Torque: } T_d = B \cdot A \cdot N \cdot I$$

Where: B = flux density, A = coil area, N = number of turns, I = current

$$\text{Deflection } \theta \propto I \quad (\text{Linear scale})$$

Advantages & Disadvantage

- Uniform (linear) scale — easy to read accurately.
- Very sensitive — suitable for measuring small currents (μA to mA).
- Not affected by external magnetic fields (due to strong permanent magnet).
- Disadvantage: Works only on DC. For AC, a rectifier must be used.

Q6b. Compare Different Types of MI Instruments & Explain Their Principles

Types of Moving Iron (MI) Instruments

Moving Iron (MI) Instrument — Repulsion Type

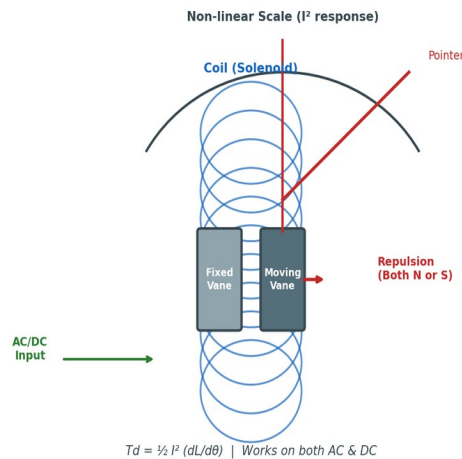


Fig 6.2 — Moving Iron (Repulsion Type) Instrument

1. Attraction Type MI

- A soft iron vane is pivoted near a coil. When current flows, the vane is attracted into the coil.
- Spring provides restoring torque. Pointer deflects proportional to I^2 .
- Scale is non-linear (cramped at lower end).

2. Repulsion Type MI (Most Common)

- Two soft iron pieces are magnetized by the coil current and repel each other.
- One vane is fixed; the other (with pointer) moves — repulsion causes deflection.

$$\text{Deflecting Torque: } T_d = \left(\frac{1}{2}\right) \times I^2 \times \left(\frac{dL}{d\theta}\right)$$

- Works on both AC and DC. Non-linear scale.

Comparison — PMMC vs Moving Iron

Parameter	PMMC Instrument	Moving Iron (MI) Instrument
Operation	DC only	AC and DC
Scale	Linear (uniform)	Non-linear (I^2 response, cramped start)
Accuracy	Very high	Moderate
Sensitivity	Very sensitive (μA range)	Less sensitive
Cost	Higher cost	Lower cost
Damping	Eddy current damping (Al former)	Air friction damping
Applications	Precise DC measurements	AC supply measurements, panel meters

UNIT 3: POWER SYSTEMS, SAFETY & ENERGY MANAGEMENT

Question 1

Q1a. Layout Diagram of Thermal Power Plant — Describe Each Part

Introduction

A thermal power plant converts heat energy (from burning coal, oil, or natural gas) into electrical energy. It is the most common type of power plant in India, providing about 60% of the total electrical power.

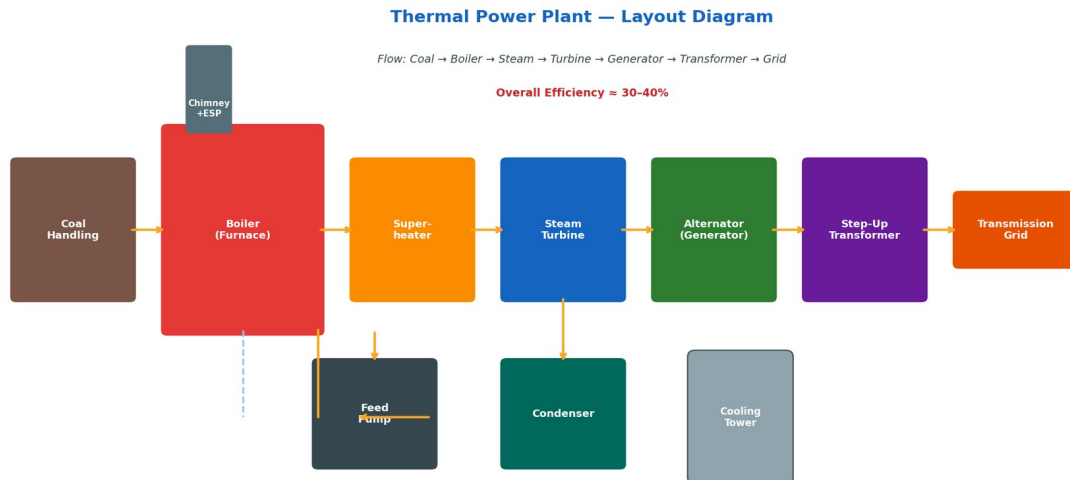


Fig 1.1 — Thermal Power Plant Layout Diagram

Components & Working

- 1. **Coal Handling Plant:** Coal is received, crushed, and stored. Crushed coal is fed to the boiler furnace by conveyors.
- 2. **Boiler (Steam Generator):** Coal is burned in the furnace. Water in tubes is heated to produce high-pressure, high-temperature steam (e.g., 500°C, 150 bar).
- 3. **Superheater:** Further heats steam beyond saturation temperature to superheated steam, improving turbine efficiency.
- 4. **Steam Turbine (HP + LP):** High-pressure superheated steam expands through turbine blades, spinning the turbine at 3000 rpm (for 50 Hz).
- 5. **Alternator (Generator):** Coupled directly to turbine shaft. Converts mechanical rotation to electrical energy (11 kV, 3-phase AC).
- 6. **Step-Up Transformer:** Steps up voltage from 11 kV to 220 kV for efficient long-distance transmission.
- 7. **Condenser:** Exhausted steam from turbine is condensed back to water using a cooling water system.
- 8. **Feed Water Pump:** Pumps condensed water back to the boiler under high pressure — completing the steam cycle.
- 9. **Cooling Tower / Cooling Water:** Circulates water to cool the condenser. Wet cooling tower is most common.
- 10. **Ash Handling Plant:** Collects fly ash and bottom ash from furnace. Fly ash is used in cement manufacturing.

- **11. Electrostatic Precipitator (ESP):** Removes fly ash particles from flue gases before they escape through chimney — pollution control.

Overall Efficiency $\approx$ 30–40% (Heat $\rightarrow$ Steam $\rightarrow$ Mechanical $\rightarrow$ Electrical)

 **NOTE:** Most energy loss in a thermal plant is in the condenser (waste heat). Combined cycle plants recover this heat and achieve 60%+ efficiency.

Q1b. Energy Consumption & Bill Calculation

Given Appliances

Appliance	Power	Qty	Time/Day	Daily Energy (kWh)
2 HP Motor	1 HP=746W $\rightarrow$ 2 $\times$ 746=1492W	1	3 hrs	1.492 $\times$ 3 = 4.476 kWh
100W Lamp	100W = 0.1 kW	2	12 hrs	0.2 $\times$ 12 = 2.400 kWh
1000W Heater	1000W = 1.0 kW	1	3 hrs	1.0 $\times$ 3 = 3.000 kWh
120W Fan	120W = 0.12 kW	2	6 hrs	0.24 $\times$ 6 = 1.440 kWh

Total Daily Energy = 4.476 + 2.400 + 3.000 + 1.440 = 11.316 kWh/day

Monthly Energy = 11.316 $\times$ 30 = 339.48 kWh ($\approx$ 339.48 Units)

Monthly Bill = 339.48 Units $\times$ ₹3 = ₹1,018.44

Question 2

Q2a. Hydro Electric Power Plant — Layout and Components

Introduction

A hydroelectric power plant converts potential energy of stored water at height into electrical energy. It uses no fuel — it is a renewable, clean energy source with very high efficiency.

Hydroelectric Power Plant — Layout Diagram

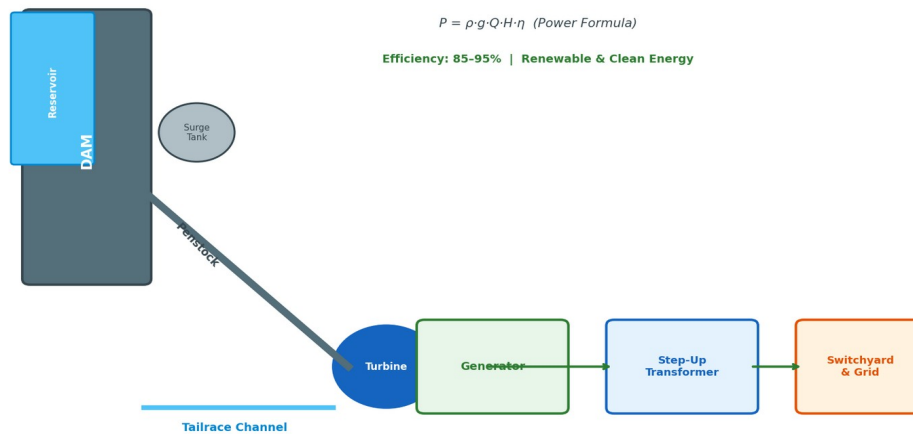


Fig 2.1 — Hydroelectric Power Plant Layout

Components & Functions

- **1. Catchment Area / Reservoir:** Large area where rain water collects behind the dam at high elevation (potential energy store).
- **2. Dam:** Built across a river to store water and create hydraulic head (height difference).
- **3. Spillway:** Safety valve — excess water is released here to prevent dam overflow.
- **4. Penstock:** Large steel pipes/tunnels carrying water at high pressure from reservoir to turbines.
- **5. Forebay / Surge Tank:** Buffer tank that absorbs pressure surges (water hammer) during load changes.
- **6. Water Turbine (Francis/Pelton/Kaplan):** High-pressure water strikes turbine blades and rotates the turbine shaft.
- **7. Alternator / Generator:** Coupled to turbine shaft, converts mechanical energy to electrical energy (AC).
- **8. Tailrace Channel:** Carries water away from turbine back to the river after passing through.
- **9. Step-Up Transformer:** Increases voltage for long-distance transmission (11 kV to 220 kV or more).
- **10. Switchyard:** Circuit breakers, isolators, and bus bars for controlling power output.

$$\text{Power Generated: } P = \rho \times g \times Q \times H \times \eta \text{ (Watts)}$$

Where: $\rho=1000 \text{ kg/m}^3$, $g=9.81 \text{ m/s}^2$, $Q=\text{flow rate (m}^3/\text{s)}$, $H=\text{head (m)}$, $\eta=\text{efficiency}$

NOTE: Hydroelectric plants have very high efficiency (85–95%) and respond quickly to load changes — ideal for peak load management.

Q2b. What is Electric Shock? Explain Safety Precautions

What is Electric Shock?

Electric shock is the physiological reaction (injury or death) caused by electric current passing through the human body. Current as low as 10 mA can cause severe muscle contractions. Current >100 mA for >1 second can be fatal.

Effects of Electric Shock by Current Level

Current Level	Effect on Human Body
1 – 10 mA	Tingling sensation, slight shock, perception threshold
10 – 25 mA	Painful contractions, unable to release (let-go threshold)
25 – 50 mA	Respiratory difficulty, possible cardiac disturbance
50 – 100 mA	Ventricular fibrillation, respiratory failure — potentially fatal
> 100 mA	Cardiac arrest, severe burns — usually fatal without immediate CPR

Safety Precautions to Avoid Electric Shock

- **1. Use Proper Insulation:** All conductors must be insulated. Check insulation regularly with a Megger.
- **2. Earthing (Grounding):** Metal bodies of equipment must be solidly earthed — fault current flows to earth safely.
- **3. Use MCB/RCCB/ELCB:** Install Residual Current Circuit Breakers (RCCB) that trip in <30 ms if leakage exceeds 30 mA.
- **4. Double Insulation:** Use Class II equipment (double-insulated tools) — two layers of insulation.
- **5. Lockout/Tagout (LOTO):** Always isolate and lock the supply switch before working on any equipment.
- **6. PPE (Personal Protective Equipment):** Use rubber gloves, rubber boots, insulated mats, and insulated tools.
- **7. Low Voltage Supply:** Use 24V or 48V DC for portable tools in wet/hazardous environments.
- **8. Do Not Work Alone:** Always have a second person present when working on live equipment.
- **9. Warning Signs:** Post danger signs near high-voltage equipment and switchgear.
- **10. Training:** All electrical workers must be trained in first aid (CPR) and electrical safety procedures.

Question 3

Q3a. Layout Diagram of Wind Power Plant — Explain Each Part

Introduction

A wind power plant converts kinetic energy of wind into electrical energy. Wind turbines are the primary device and can be installed onshore or offshore. Power in wind: $P = \frac{1}{2}\rho A v^3$

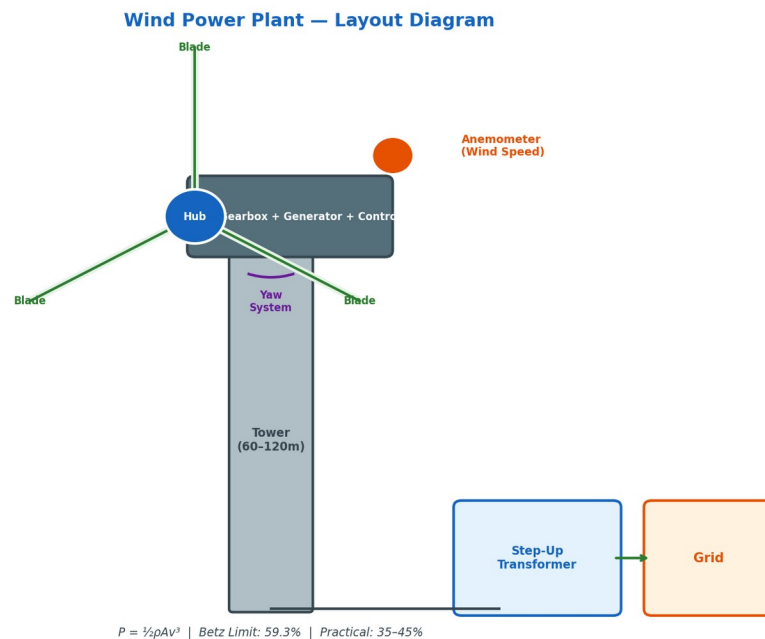


Fig 3.1 — Wind Power Plant Layout Diagram

Components & Functions

- **1. Wind Resource / Site:** Location with consistent high wind speed (>7 m/s average). Hilltops, coastal areas are ideal.
- **2. Anemometer:** Measures wind speed. Controls turbine operation — starts at cut-in speed (~3–4 m/s), stops at cut-out (~25 m/s).
- **3. Wind Vane:** Measures wind direction. Works with yaw motor to orient nacelle into the wind.
- **4. Rotor Blades (3 blades):** Aerodynamically shaped blades (30–80 m long, glass-fibre reinforced plastic) capture wind energy.
- **5. Hub:** Connects blades to main shaft. Has pitch control to adjust blade angle based on wind speed.
- **6. Gearbox:** Steps up low rotor speed (~15–20 rpm) to high generator speed (~1500 rpm).
- **7. Generator:** Converts mechanical rotation to AC electricity. Induction or synchronous type.
- **8. Nacelle:** Weatherproof housing on top of tower containing gearbox, generator, brake, and control systems.
- **9. Yaw System:** Rotates entire nacelle to face into wind (active yaw control by electric motor).
- **10. Tower (Steel/Concrete):** Tall tubular tower (60–120 m) supports nacelle and rotor. Taller = stronger, more consistent wind.
- **11. Transformer:** Steps up voltage from generator (690V) to grid level (11 kV or 33 kV).
- **12. SCADA System:** Supervisory control monitors performance, detects faults, controls pitch and yaw remotely.

$$P = \left(\frac{1}{2} \right) \times \rho \times A \times v^3 \quad (\text{Watts})$$

 **NOTE:** Maximum theoretical efficiency of wind turbine = 59.3% (Betz Limit). Practical efficiency is 35–45%.

Q3b. Explain the Operation of Fuse and MCB

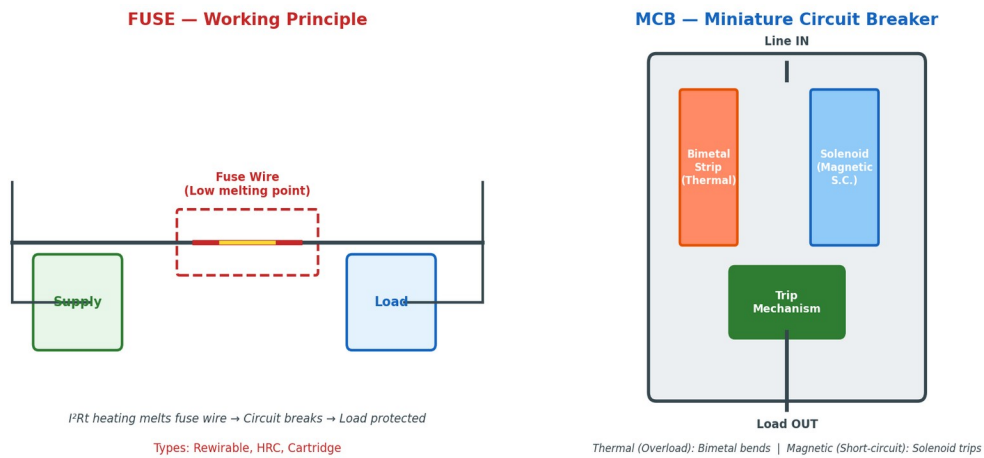


Fig 3.2 — Fuse & MCB Construction and Working

FUSE — Definition & Working

A fuse is a short length of thin wire (of low-melting-point material like tin-lead alloy) connected in series with a circuit to protect it from overcurrent.

Working Principle of Fuse

- When current exceeds the rated value, the fuse wire heats up due to I^2Rt joule heating.
- The fuse wire melts and breaks the circuit, disconnecting the load from the supply.
- Fuses are one-time devices — they must be replaced after every operation.
- **Types:** Rewirable fuse, HRC (High Rupturing Capacity) fuse, Cartridge fuse

MCB — Miniature Circuit Breaker

An MCB is an electromechanical device that automatically switches OFF the electrical circuit during abnormal conditions (overload or short circuit) and can be reset manually.

Working Principle of MCB

- **Thermal Protection (Overload):** A bimetallic strip bends when heated by excess current and trips the breaker.
- **Magnetic Protection (Short Circuit):** A solenoid creates a strong magnetic field with high fault current, which instantly repels the trip mechanism.

Parameter	Fuse	MCB (Miniature Circuit Breaker)
Reusability	One-time use (replace after operation)	Reusable — reset after fault cleared
Response Speed	Slower (thermal melting)	Very fast (<0.1 ms for short circuit)
Cost	Very low initial cost	Higher initial cost
Construction	Simple wire/element	Has moving mechanical trip mechanism
Protection Type	Overcurrent only	Both overload and short circuit

Parameter	Fuse	MCB (Miniature Circuit Breaker)
Resetting	Cannot be reset — must replace	Can be reset manually
Safety	Fails safe by melting (isolates)	Provides clear ON/OFF status indication

Question 4

Q4a. Layout Diagram of Solar Power Plant — Explain Each Component

Introduction

A solar power plant (photovoltaic system) converts sunlight directly into electricity using the photovoltaic effect. It is clean, silent, and renewable with zero fuel cost and 25+ years operational life.

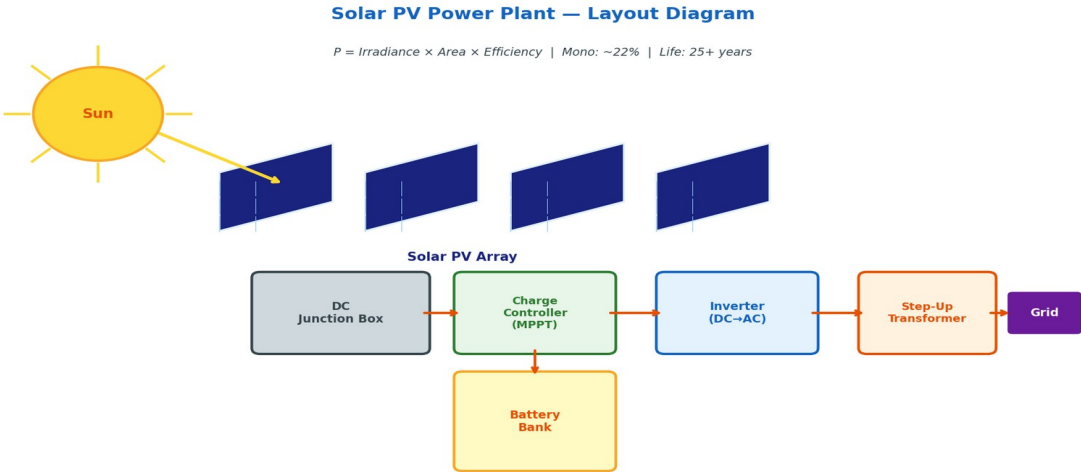


Fig 4.1 — Solar PV Power Plant Layout

Components & Functions

- **1. Solar PV Panels:** Arrays of solar cells (silicon p-n junction) that generate DC electricity when sunlight strikes. Each cell: ~0.5–0.6V. Cells → Modules → Arrays.
- **2. Solar Tracking System (Optional):** Mechanical system that rotates panels to follow the sun (single/dual-axis tracking) for 20–40% more energy.
- **3. DC Junction Box:** Combines output from multiple strings of panels. Has bypass diodes and string fuses for protection.
- **4. Charge Controller (MPPT):** For off-grid systems — regulates battery charging. MPPT type extracts maximum power from panels.
- **5. Battery Bank (Off-grid):** Stores DC energy produced during daytime for use at night or during cloudy periods. Lead-acid or Li-ion batteries.
- **6. Inverter (Grid-Tied/Off-Grid):** Converts DC electricity from panels/batteries to AC electricity (230V, 50 Hz) for homes and grid.

- **7. Step-Up Transformer (Large plants):** Steps up voltage from inverter output to grid level (11 kV, 33 kV, or 132 kV) for bulk transmission.
- **8. AC Distribution Board / Switchgear:** Circuit breakers, metering, protection relays for AC side management.
- **9. Net Meter:** Bidirectional meter for grid-tied systems — records power sent to grid and power consumed from grid.
- **10. Monitoring System (SCADA):** Continuously monitors panel temperature, irradiance, voltage, current, energy yield, and faults.

Power Output: $P = \text{Irradiance (W/m}^2\text{)} \times \text{Panel Area (m}^2\text{)} \times \text{Panel Efficiency (\%)}\text{'}$

Energy per day = $P \times \text{Peak Sun Hours (PSH)}$ [kWh/day]

NOTE: PV cell efficiency: Monocrystalline ~22%, Polycrystalline ~17–18%, Thin Film ~10–13%. One-time investment, 25+ year life, zero running cost.

Q4b. Explain Different Types of Earthing

What is Earthing?

Earthing (or grounding) is the process of connecting the non-current-carrying metal parts of electrical equipment to the earth through a low-resistance conductor. It protects people and equipment from electric shock caused by insulation failure.

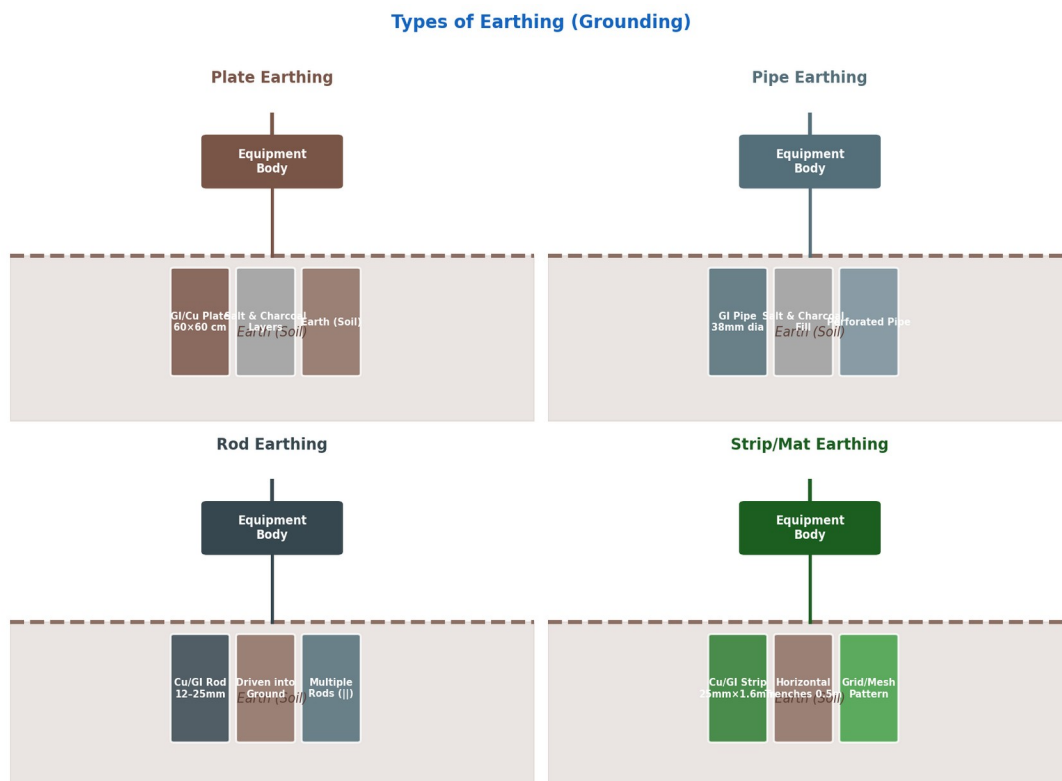


Fig 4.2 — Types of Earthing

1. Plate Earthing

- A copper plate (60 cm × 60 cm × 3.18 mm) or GI plate (60 cm × 60 cm × 6.35 mm) is buried vertically at depth ≥ 3 metres.
- Surrounded by alternate layers of coal (charcoal) and salt to keep earth resistance low.
- An earthing conductor connects the plate to the equipment body via a GI pipe.
- Water is poured periodically through the pipe to keep surrounding soil moist.
- Used for: large buildings, transformer neutrals, distribution systems.

2. Pipe Earthing

- A GI pipe (38 mm dia, 2 metres long) with perforations is buried vertically in the earth.
- Pipe is surrounded with salt and charcoal mixture to maintain low resistance.
- Most common type of earthing in India. Cheaper than plate earthing.
- Used for: residential buildings, small commercial establishments.

3. Rod Earthing

- Copper or GI rods (12–25 mm diameter) are driven vertically into the earth.
- Multiple rods may be connected in parallel to achieve required earth resistance ($<1\ \Omega$ for power stations, $<5\ \Omega$ for domestic).
- Used where rocky soil prevents deep burial — rods are hammered in.

4. Strip Earthing (Mat Earthing)

- Copper or GI strips (25 mm × 1.6 mm) are buried horizontally in trenches about 0.5 m deep.
- Used where very low earth resistance is required — power stations, substations, hospitals.
- GI mesh (earthing grid) is used in switchyards for step potential protection.

Type	Electrode	Depth	Best For	Earth Resistance
Plate Earthing	Cu/GI plate 60×60 cm	≥ 3 m vertical	Transformers, distribution	$< 1\ \Omega$
Pipe Earthing	GI pipe 38 mm dia	2 m vertical	Residential, commercial	$< 5\ \Omega$
Rod Earthing	Cu/GI rod 12–25 mm	Driven deep	Rocky soil areas	$< 5\ \Omega$
Strip Earthing	Cu/GI strip 25mm×1.6mm	0.5 m horizontal	Power stations, substations	$< 1\ \Omega$

 **NOTE:** All earthing types must achieve earth resistance $<1\Omega$ for power stations, $<2\Omega$ for substations, and $<5\Omega$ for domestic installations as per IS 3043.